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Arinze Elenasulu

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Python Backend Engineer | AI & Automation

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Professional Summary
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I have been writing Python since 2020 and I build software by sitting close to the problem. I built the backend for a pharmacy operations platform by sitting with pharmacy staff, understanding their manual processes, and turning paper workflows into software they use every day. I run a production AI agent on cloud infrastructure and built governance tooling for AI coding agents published on PyPI. I work best when I am talking to the people who will actually use what I build, figuring out what matters, and shipping fast. I am looking for a forward deployed role where technical depth meets real customer problems.

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Core Skills

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Backend Engineering: Python (primary, since 2020), FastAPI, SQLAlchemy, Alembic, REST API design, async patterns, PostgreSQL, schema design, query optimisation
AI & Automation: Agentic workflow orchestration, MCP tooling, LLM integration, RAG pipeline design, structured output design, AI governance and project memory systems
Infrastructure & Reliability: Docker, Linux, cloud VPS administration, structured logging, input validation, defensive backend design, error handling at the system boundary
Testing & Quality: Advanced pytest (fixtures, parametrisation, integration coverage), structured test design, API test automation
Additional Languages & Stacks: Java, Spring Framework, C++, C, JavaScript/TypeScript, MERN stack, MEAN stack, Flutter/Dart (from formal training. Python is my strongest)

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Experience
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Vigilis, Software Engineer
August 2024 – February 2026 · Lagos, Nigeria
Built and maintained the backend for a pharmacy operations platform handling sales, inventory, payments, and reporting across multiple locations. System used day-to-day by pharmacy staff.

- Designed the full data model and API layer (FastAPI, PostgreSQL, SQLAlchemy) with role-based access, JWT auth, and Alembic migrations that kept the database evolving safely as features shipped.
- Sat with pharmacy staff directly to understand their manual processes. Turned paper workflows and mental checklists into software that cut daily admin time and reduced stock discrepancies.
- Built the invoice lifecycle: create, add line items, finalise with automatic stock deduction, cancel with stock reversal. Every state transition validated and audited.
- Added structured logging, input validation, and business rule enforcement that caught errors at the API boundary before they could corrupt data downstream.
- Owned the backend end to end: feature planning, database design, deployment, monitoring, and production bug fixes. No handoff.

Aptech Computer Education, Software Engineer Intern

September 2022 – August 2024 · Lagos, Nigeria

Completed the Advanced Diploma in Software Engineering (ADSE Java track), a structured two-year programme combining classroom training with hands-on project work.

- Covered the full software engineering stack across four semesters: Linux system administration, Python, C++, C, Java, Spring Framework, Enterprise Java Beans (EJB), Flutter/Dart for mobile, and both MERN and MEAN stacks for full-stack web development.
- Built multiple projects across different technology stacks as part of the programme curriculum, gaining practical exposure to how different languages and frameworks solve similar problems.
- Developed strong fundamentals in object-oriented programming, database design, and software architecture that carried into every role since.
- Python became my primary language during this period and has remained so. I have been studying and building with it continuously since 2020.

Bucca Hut, AI and Backend Engineer (Contract, part-time)

February 2025 – May 2025

Built a data analysis tool for a food business, connecting LLM outputs to real operational decisions. Designed the backend API bridging AI processing with the existing workflows.

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Selected Projects

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FRAME, Typed Project Memory Format for AI Agents

Python, Markdown/YAML, Multi-Agent

github.com/haxsysgit/FRAME

Designed a portable standard that gives AI coding agents structured project context across sessions.

- Five-part memory model (Facts, Rules, Acts, Map, Expect) capturing project truth, agent rules, session history, module relationships, and expected outcomes.
- Reduces repeated prompting, context rot, and full-repo rescans across agent sessions.
- Designed as a portable standard: any AI agent reads the same project representation.
- Published docs, learn materials, and frame-python implementation track on GitHub.
- Separation of concerns: FRAME is the contract, implementations earn alignment with it.

Haxaml, AI Agent Governance & Project Memory

Python, MCP, FRAME

github.com/haxsysgit/haxaml · pypi.org/project/haxaml

Built and published a governance runtime that enforces project rules for AI coding agents.

- Governance runtime enforcing FRAME correctness contract, blocks AI-generated code until mechanical checks pass.
- Five-stage pipeline: Admission → Scoped Context → Agent Work → Verification → Recording.

- MCP server so AI coding tools can read and write to FRAME files during a session.
- Local dashboard for inspecting agent session history and project state.
- Published on PyPI with documentation, setup flow, and adoption guides.

Pharmax, Pharmacy Management Platform

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic

Built the backend for a pharmacy operations platform handling product inventory, sales, invoicing, and reporting.

- Products CRUD with multi-unit pricing (tablets, packs, strips).
- Full invoice lifecycle: draft to finalise (auto stock deduction) to cancel (stock reversal).
- Stock adjustments with audit trail and reorder-level tracking.
- Role-based access with JWT authentication.
- Comprehensive pytest suite covering endpoints, business logic, and edge cases. Sat with pharmacy staff to understand manual processes and turned them into software.

HaxJobs, Automated Job Search & Application Platform

Python, Playwright, React, TypeScript, Hermes Agent

github.com/haxsysgit/haxjobs

Built an end-to-end automated job discovery and application platform powered by AI agent infrastructure.

- Multi-platform job discovery across 95+ companies (Lever, Ashby, Greenhouse APIs).
- Hermes-powered job evaluation pipeline scoring fit to user profile.
- 5-level automated application pack generation with CV, cover letter, Q&A.
- Web dashboard for pipeline monitoring with real-time job status views.
- Zero-token discovery via system-level cron scheduling, independent of LLM.

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Education

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BSc Information Technology , Middlesex University London (Expected June 2026)

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Advanced Diploma in Software Engineering (ADSE Java) , Aptech Computer Education, Lagos (September 2022 – August 2024)

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